

PnM DPR & Machinery Management System

PRODUCT DOCUMENTATION

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RVR Projects

Plant & Machinery (PnM) Division

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1. Executive Summary

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The PnM DPR & Machinery Management System is a web-based enterprise application developed for RVR Projects to digitise and streamline Plant & Machinery operations across construction sites. The system replaces manual daily log sheets, paper-based fuel receipts, and Excel-based attendance records with a unified digital platform.

Core capabilities include: machine utilisation tracking via daily shift-wise DPR entries, fuel consumption monitoring, preventive and corrective maintenance logging, operator attendance and payroll, hire work order management with multi-level approval, spare parts inventory, and breakdown incident tracking.

Built on a modern technology stack (React + Express.js + PostgreSQL), the system supports multi-project deployment with role-based access control. Each stakeholder — from field operator to senior management — sees only the data relevant to their role, ensuring data integrity and accountability.

The application is designed for field use: mobile-responsive layouts, one-click Excel and PDF report generation, and bulk import via downloadable Excel templates for quick onboarding of large fleets and teams.

2. System Architecture

The system follows a standard three-tier web architecture: React SPA (frontend), Express REST API (backend), and PostgreSQL database. The frontend is served as a static build; the backend exposes a JSON REST API secured with JWT. All persistent data is stored in PostgreSQL with schema changes managed via numbered migration files.

2.1 Frontend

React 18 single-page app built with Vite. Client-side routing via React Router v6. Tailwind CSS for responsive styling. Axios for API communication. Client-side PDF generation via jsPDF + autotable; Excel via SheetJS.

2.2 Backend

Node.js Express REST API. JWT Bearer token auth with `requireAuth` and `requireAdmin` middleware. Rate limiting: 300 req/15min globally, 15 req/15min on login. Helmet security headers. Configurable CORS. Health check at GET / health.

2.3 Database

PostgreSQL 14+ with 23 sequential numbered SQL migration files in `backend/migrations/`. Raw SQL via `node-postgres` — no ORM. Migrations are additive and run once; the runner tracks applied files by filename.

3. User Roles & Permissions

Three roles with progressively restricted access:

- Admin — Full system access: all modules, CRUD, bulk imports, user/project management, payroll, hard deletes.
- Site Incharge — Project-level: DPR, fuel, attendance, breakdown, spare parts. View-only hire WOs. No payroll or admin functions.
- Operator — Entry-only: DPR entry and dashboard. Cannot edit or delete any record.

Feature / Module	Admin	Site Incharge	Operator
Login & Profile	Yes	Yes	Yes
My Dashboard (Fleet Status)	Yes	Yes	Yes
DPR Entry (own project)	Yes	Yes	Yes
Edit / Delete DPR Entries	Yes	No	No
DPR Download (PDF)	Yes	Yes	No
Fuel Entry	Yes	Yes	Yes
Service Entry	Yes	Yes	No
Utilisation & Summary Reports	Yes	Yes	Yes
Add / Edit Machines	Yes	If can_add_assets	No
Bulk Import (Machines/Users)	Yes	No	No
Transfer Machines	Yes	No	No
User Management	Yes	No	No
Project Management	Yes	No	No
Equipment Types	Yes	No	No
HR: Operators & Attendance	Yes	Yes	No
HR: Payroll Generation	Yes	No	No
Asset Register Download	Yes	Yes	No
Hire Work Orders (View)	Yes	Yes	No
Hire Work Orders (Create/Approve)	Yes	No	No
Breakdown Management	Yes	Yes	No
Spare Parts Inventory	Yes	Yes	No
Delete Any Record	Yes	No	No

4.1 Authentication & User Management

Purpose

Secure login and role-based access for all system users.

Key Features

- JWT-based authentication — 7-day token expiry, stored in localStorage
- Password hashing using bcryptjs (cost factor 10)
- Three user roles: Admin, Site Incharge, Operator
- Per-user multi-project assignment via project_codes array
- Bulk user import via downloadable Excel template
- Profile fields: mobile, email, department, joining date, photo URL
- Last login timestamp tracking for audit trail
- "Can Add Assets" permission flag for elevated non-admin users
- Rate limiting on login endpoint (15 requests per 15 minutes per IP)

Workflow

User logs in with username/password !' JWT issued by server !' Role-based routes unlocked in frontend !' Subsequent API calls carry Bearer token in Authorization header

4.2 Project Management

Purpose

Central project registry linking all operational data across the system.

Key Features

- Project code, name, and site address fields
- Project-based data segregation across all modules
- Users and operators assigned to one or more specific projects
- Multi-project access for Site Incharge role
- All DPR, fuel, HR, and inventory data tagged to a project

Workflow

Admin creates project with code and address !' Assigns users and operators !' All subsequent data entries (DPR, fuel, attendance, breakdown) are tagged to the project

4.3 Fleet Management (Machine Registry)

Purpose

Central registry of all plant and machinery assets across all projects.

Key Features

- Asset categorisation: Own vs Hire
- Equipment type classification (Dozer, Excavator, Crane, Generator, Compactor, etc.)
- Machine status: Active, Idle, Breakdown, Not Deployed
- Readings basis: Hours or Kilometre
- Shift assignment: Day / Night / Full Day
- Fuel parameters: min/max km reading bands for fuel validation
- Asset code, serial number, registration number, capacity fields
- Rate monthly field for hire cost tracking
- Machine transfer between projects with transfer date logging
- Bulk import via downloadable Excel template
- Hard delete capability (Admin only)
- Fleet status summary (pie chart on My Dashboard)

Workflow

Equipment Type defined !' Machine registered with type, ownership, shift !' Assigned to project !' Status auto-reflected via daily DPR entries !' Transferred with date when relocated to another project

4.4 Daily Performance Report (DPR) Entry

Purpose

Digital shift-wise daily log capturing machine utilisation for the entire fleet.

Key Features

- Opening and closing meter reading per shift (supports dual-shift Day/Night)
- Auto-calculation of work done (hours or km)
- Fuel quantity consumed per shift entry
- Breakdown flag with description field
- Work description and activity remarks
- Previous closing reading auto-populated from last entry
- Monthly calendar history view per machine
- DPR completion status indicators per machine per day
- Operator entries are immutable after submission
- Admin can edit or delete any entry
- Admin DPR view with full filter and edit/delete capability

Workflow

Operator selects machine and shift date !' Opening/closing readings entered !' Breakdown flagged if applicable !' Entry submitted !' Data feeds Utilisation report, Daily Summary, and DPR Download

4.5 Fuel Management

Purpose

Track fuel receipts and issues to machines across all projects.

Key Features

- Fuel types: Diesel, Petrol, CNG
- Quantity (litres) and rate per litre capture
- Linked to machine, project, and shift date
- Cumulative consumption aggregated in utilisation reports
- Admin-only deletion

Workflow

Fuel received at site !' Entry logged with quantity, rate, fuel type, machine !' Monthly consumption aggregated in reports and dashboards

4.6 Service & Maintenance

Purpose

Track all preventive and corrective maintenance activities.

Key Features

- Service types: Preventive, Corrective, Breakdown Service
- Mechanic name and workshop/garage field
- Meter reading at service and next service due reading
- Parts replaced description
- Service cost capture

Workflow

Maintenance completed !' Service entry logged with readings and cost !' Next service due reading tracked !' Historical service records available per machine

4.7 HR Module — Operators, Attendance & Payroll

Purpose

End-to-end management of field workforce from onboarding to salary disbursement.

Key Features

- Operator master: designation, mobile, licence number, joining date, daily wage, active/inactive status
- Daily attendance statuses: Present, Absent, Half Day, On Leave, Holiday
- Shift types per attendance: Day, Night, Full Day
- OT (overtime) hours capture per attendance record
- Payroll run generation per project per pay period from attendance data
- Payroll items: present days, absent days, OT hours, basic pay, OT pay, deductions, net pay
- Payroll status workflow: Draft !' Approved !' Paid
- Designation master for classification

Workflow

Operators onboarded with wage rates !' Daily attendance marked per shift !' End of period: Payroll Run generated from attendance totals !' Admin reviews and sets status to Approved !' Marks as Paid after disbursement

4.8 Asset Register

Purpose

Structured asset register for audit, insurance tracking, and compliance.

Key Features

- Three register views: Own Measurable, Own Non-Measurable, Hire Assets
- Excel and PDF download for each category
- Asset code, model, manufacturer, year of manufacture
- Insurance expiry, PUC expiry, fitness certificate date fields
- Bulk upload via Excel template

Workflow

Machines registered in Fleet Management !' Automatically categorised into Asset Register views based on ownership (Own/Hire) and equipment type measurement category (Measurable/Non-Measurable)

4.9 Hire Work Orders

Purpose

End-to-end procurement management for hired equipment and services.

Key Features

- Work order with multiple equipment line items
- Vendor master: name, GST number, PAN, contact, bank details
- Real-time GST number verification via external API with local format validation fallback
- Multi-level approval workflow: Draft !' Submitted !' L1 Approved !' Approved / Rejected
- Work order renewal with link to original order preserved
- Vendor offer fields for commercial comparison before award
- Rate monthly and estimated total cost calculation per WO

Workflow

Vendor created with GST verification !' Work Order drafted with equipment line items and rates !' Submitted for L1 review !' Final approval !' Equipment mobilised to project !' Renewed at contract expiry

4.10 Breakdown & Incident Management Aggregates Breakdown Projects

Purpose

Track machine breakdowns from occurrence through to resolution.

Key Features

- Incident description, root cause, and corrective action taken fields
- Downtime hours and repair cost capture
- Status lifecycle: Open !' In Progress !' Resolved
- Breakdown summary report: incidents, total downtime, repair cost by machine and project
- Linked to DPR breakdown flags

Workflow

Breakdown flagged in DPR entry !' Admin creates formal Incident record !' Repair work assigned and tracked !' Status progressed through In Progress !' Resolved with downtime hours and cost logged

4.11 Spare Parts Inventory

Purpose

Track spare parts stock levels and movements across projects.

Key Features

- Transaction types: Receipt (stock in), Issue (stock out), Return
- Item name, part number, unit of measure, quantity, unit cost fields
- Running stock balance / stock summary view
- Linked to machines, projects, and service/breakdown jobs

Workflow

Parts received !' Stock incremented via Receipt transaction !' Issued against machine service or breakdown !' Returns recorded for surplus !' Stock summary shows live balance per item

4.12 Reports & Analytics

Purpose

Aggregate operational data into actionable management reports.

Key Features

- Utilisation Report: days reported, total work hours, average utilisation %, fuel consumed — per machine; date-range and project filters
- Daily Summary: total machines on site, aggregate hours, overall utilisation rate
- DPR Download Modal: date-range filtered PDF export of formatted daily reports
- Breakdown Summary: incident count, total downtime hours, total repair cost by machine and project
- Machine Download: filterable asset list export in Excel and PDF
- CSV export available on all major data listing views
- My Dashboard: fleet status pie chart (Active / Idle / Breakdown / Not Deployed)

5. Technology Stack

Layer	Technology	Version	Purpose
Frontend	React	18.3.1	UI component framework
Frontend	Vite	5.3.1	Build tool & dev server
Frontend	Tailwind CSS	3.4.4	Utility-first CSS styling
Frontend	React Router DOM	6.24.0	Client-side SPA routing
Frontend	Axios	1.7.2	HTTP client for API calls
Frontend	jsPDF + autotable	4.2.1 / 5.0...	Client-side PDF generation
Frontend	XLSX (SheetJS)	0.18.5	Excel file read/write
Frontend	date-fns	3.6.0	Date formatting & calculation
Frontend	Lucide React	0.400.0	SVG icon library
Backend	Node.js	20+	JavaScript server runtime
Backend	Express	4.19.2	HTTP web framework
Backend	PostgreSQL	14+	Primary relational database
Backend	pg (node-postgres)	8.11.5	PostgreSQL client driver
Backend	jsonwebtoken	9.0.2	JWT creation & validation
Backend	bcryptjs	2.4.3	Password hashing (bcrypt)
Backend	Helmet	7.1.0	HTTP security headers
Backend	express-rate-limit	7.3.1	API rate limiting
Backend	dotenv	16.4.5	Environment variable loading
Backend	cors	2.8.5	Cross-Origin Resource Sharing
Backend	nodemon	3.1.4	Dev server hot-reload

6. Database Schema Overview

All primary database tables, their purpose, and key fields. Full DDL is in backend/migrations/ (23 migration files).

Table	Purpose	Key Fields
projects	Project master data	id, code, name, address, created_at
users	All application users	id, username, role, project_codes[], can_add...
equipment_types	Equipment type master	id, name, category, unit (Hours/Km)
machines	Full fleet registry	id, asset_code, serial_no, type_id, project_...
dpr_entries	Shift-wise daily perfor...	id, machine_id, project_id, date, shift, ope...
fuel_entries	Fuel receipt/issue log	id, machine_id, project_id, date, fuel_type,...
service_entries	Maintenance records	id, machine_id, service_type, date, mechanic...
operators	Field workforce registry	id, emp_id, name, designation_id, mobile, li...
attendance	Daily attendance log	id, operator_id, project_id, date, shift, st...
payroll_runs	Payroll batch runs	id, project_id, period_from, period_to, stat...
payroll_items	Per-operator payroll li...	id, run_id, operator_id, present_days, ot_ho...
breakdown_inciden...	Machine breakdown events	id, machine_id, project_id, date, descriptio...
spare_transactions	Spare parts movements	id, machine_id, project_id, txn_type (Receip...
hire_vendors	Hire vendor master	id, name, gst_no, pan_no, contact_person, mo...
hire_work_orders	Hire procurement WOs	id, vendor_id, project_id, wo_number, status...
hire_wo_items	WO equipment line items	id, wo_id, machine_id, description, qty, rat...

7. API Reference

All endpoints prefixed /api. Auth: Bearer <JWT> in Authorization header. Rate limit: 300 req/15min per IP; 15 req/15min on login; 20 req/min on GST verify.

Authentication

Method	Endpoint	Description	Access
POST	/api/auth/login	Login — returns JWT token	Public
GET	/api/auth/me	Get current user profile	User
PUT	/api/auth/me	Update own profile	User

Projects

Method	Endpoint	Description	Access
GET	/api/projects	List all projects	User
POST	/api/projects	Create project	Admin
PUT	/api/projects/:id	Update project	Admin
DELETE	/api/projects/:id	Delete project	Admin

Machines (Fleet)

Method	Endpoint	Description	Access
GET	/api/machines	List machines with filters	User
GET	/api/machines/fleet-summary	Status count summary	User
POST	/api/machines	Create machine	Admin/Ca...
POST	/api/machines/bulk	Bulk import from Excel	Admin
PUT	/api/machines/:id	Update machine	Admin
PUT	/api/machines/:id/transfer	Transfer to project	Admin
DELETE	/api/machines/:id	Soft delete	Admin
DELETE	/api/machines/:id/hard	Hard delete	Admin

DPR Entries

Method	Endpoint	Description	Access
GET	/api/entries	List entries with filters	User
POST	/api/entries	Create DPR entry	User
PUT	/api/entries/:id	Edit entry	Admin
DELETE	/api/entries/:id	Delete entry	Admin
GET	/api/entries/dpr-status	Daily completion status	User
GET	/api/entries/monthly-status	Monthly calendar status	User

Method	Endpoint	Description	Access
GET	/api/entries/previous-closing	Last closing reading	User

Fuel, Service, Breakdown

Method	Endpoint	Description	Access
GET/POST	/api/fuel	Fuel entries list / create	User
DELETE	/api/fuel/:id	Delete fuel entry	Admin
GET/POST	/api/service	Service entries list / create	User
GET/POST	/api/breakdown	Breakdown incidents	User/Adm...
PATCH	/api/breakdown/:id	Update breakdown status	Admin

HR Module

Method	Endpoint	Description	Access
GET/POST/PUT/D...	/api/operators	Operator CRUD	Admin
GET/POST/DELETE	/api/attendance	Attendance CRUD	User/Adm...
GET	/api/payroll	List payroll runs	Admin
GET	/api/payroll/:id/items	Payroll run items	Admin
POST	/api/payroll/generate	Generate payroll run	Admin
PATCH	/api/payroll/:id	Update payroll status	Admin

Hire Work Orders

Method	Endpoint	Description	Access
GET/POST	/api/hire	List / create WOs	Admin
GET	/api/hire/:id	Get WO detail	Admin
PUT/DELETE	/api/hire/:id	Update / delete WO	Admin
PATCH	/api/hire/:id/submit	Submit for approval	Admin
PATCH	/api/hire/:id/approve-l1	Level-1 approval	Admin
PATCH	/api/hire/:id/approve	Final approval	Admin
PATCH	/api/hire/:id/reject	Reject WO	Admin
POST	/api/hire/:id/renew	Renew work order	Admin
GET/POST/PUT/D...	/api/hire/vendors	Vendor CRUD	Admin

Reports, GST & Others

Method	Endpoint	Description	Access
GET	/api/reports/utilization	Machine utilisation analytics	User
GET	/api/reports/summary	Daily operational summary	User

Method	Endpoint	Description	Access
GET	/api/reports/breakdown-summary	Breakdown incident summary	User
POST	/api/gst/verify	Verify GST number (20/min limi...	User
GET	/api/gst/validate	Local GST format validation	User
GET/POST/DELETE	/api/spare-parts	Spare parts CRUD	User/Adm...
GET	/health	API health check	Public

Prerequisites

- Node.js v20 or higher (check: `node --version`)
- PostgreSQL v14 or higher (check: `psql --version`)
- npm v10 or higher (check: `npm --version`)

Backend Setup

1. `cd backend`
2. `npm install`
3. Copy `.env.example` to `.env` and fill in all values (see Environment Variables below)
4. Run database migrations: `node migrations/run.js`
5. (Optional) Seed initial data: `node seeds/seed.js`
6. Start server: `node server.js` OR `nodemon server.js` (for development)

Frontend Setup

1. `cd frontend`
2. `npm install`
3. Create `frontend/.env` with: `VITE_API_URL=http://localhost:3000`
4. Development server: `npm run dev` (opens at `http://localhost:5173`)
5. Production build: `npm run build` (outputs to `frontend/dist/`)
6. Deploy: serve the `dist/` folder via Nginx, Vercel, or any static host

Environment Variables

Variable	Description	Example / Notes
DATABASE_URL	PostgreSQL connection string	<code>postgres://user:pass@host:5432/...</code>
JWT_SECRET	Strong random secret (min 32 cha...	Generate with: <code>openssl rand -he...</code>
PORT	Backend server port	3000
CORS_ORIGIN	Allowed frontend origin	<code>http://localhost:5173</code>
GST_API_KEY	(Optional) External GST API key	From GST verification portal pr...

9. Version History

Version	Period	Release Notes
0.1.0	2024 Q3	Initial DPR entry form, basic dashboard, JWT authentication
0.2.0	2024 Q4	Fleet management module, fuel entries, service/maintenance entries
0.3.0	2025 Q1	HR module: operator master, daily attendance tracking, payroll ru...
0.4.0	2025 Q2	Asset Register (Own Measurable/Non-Measurable/Hire); bulk Excel i...
0.5.0	2025 Q3	Hire Work Orders with multi-level approval workflow; GST number v...
0.6.0	2025 Q4	Breakdown incident tracking with status workflow; spare parts inv...
0.7.0	2026 Q1	Machine status fields (Active/Idle/Breakdown/Not Deployed); proje...
1.0.0	12 May 2026	Production release: complete module suite, role-based access cont...